

Prompt Framing Lab Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	prompt-framing-lab

Big idea: A prompt is not just a request; it frames what kind of knowledge output is likely.

Student-Facing Prompt

Inspect Prompt Framing Lab Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Prompt Framing Lab Stimulus A	Student-facing stimulus 1: Neutral prompt: 'Explain whether homework improves learning.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Prompt Framing Lab Stimulus B	Student-facing stimulus 2: Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Prompt Framing Lab Reveal / Twist	Student-facing stimulus 3: Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A prompt is not just a request; it frames what kind of knowledge output is likely.

Actual Lesson Questions

#	Question for students
1	After inspecting Prompt Framing Lab Stimulus A, what is your first knowledge claim?
2	Which exact detail from Prompt Framing Lab Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Who is responsible for knowledge produced through a prompt?

Ready-to-Use Examples

- Neutral prompt: 'Explain whether homework improves learning.'
- Leading prompt: 'Explain why homework clearly improves learning and list supporting evidence.'

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Activity Steps

1. Give groups the same knowledge task with different prompt wordings.
2. Students predict how their prompt might shape the answer.
3. Groups compare generated or mock outputs for tone, evidence, uncertainty, and omissions.
4. Students revise one prompt to improve reliability and transparency.
5. Debrief how human framing and automated generation interact.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Who is responsible for knowledge produced through a prompt?
- Can a better question make a machine answer more reliable?
- When does a prompt smuggle in an assumption?

Knowledge Questions

- Who is responsible for knowledge produced through a prompt?
- How does question design shape technological knowledge?
- Can prompts improve justification or merely improve fluency?

Transfer Example

For example, run this activity with two ai prompts that ask for confident advice versus cautious analysis. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...